

Vehicle Service Operations Analytics & Process Optimization

Complete Project Specification | Business Intelligence + Business Analysis

Purpose: Build an end-to-end analytics solution that transforms automotive service operations data into business insights, identifies process bottlenecks, and recommends measurable improvements.

Audience: Recruiters, hiring managers, project reviewers, and developers building the portfolio project.

Item	Specification
Project type	Business Intelligence + Business Analysis + Process Optimization
Domain	Automotive vehicle servicing / after-sales operations
Primary tools	Python, SQL, Power BI, Excel
Database	PostgreSQL or MySQL
Dataset	10,000–25,000 synthetic service records
Main output	Power BI dashboard + analysis + requirements/process documentation
Difficulty	Beginner to intermediate
Target duration	2–3 weeks for a polished version

1. Business Problem

A multi-location automotive service organization has data covering appointments, vehicles, technicians, parts, costs, completion times, cancellations, and customer feedback. Management needs a reliable way to understand operational performance and determine where process improvements are required.

The project must answer these management questions:

- How much revenue, cost, and profit is generated, and how does performance change over time?
- Which service centers, technicians, vehicle categories, and service types perform best or worst?
- Where do delays, cancellations, cost overruns, and customer dissatisfaction occur?
- What operational factors are associated with poor turnaround time and service performance?
- What practical process changes should management implement based on evidence?

The final outcome must be actionable recommendations, not simply charts.

2. Objectives

- Create a realistic synthetic automotive service dataset.
- Store and query the data using SQL.
- Clean, validate, and analyze the data using Python.
- Calculate operational, financial, and customer-experience KPIs.
- Build an interactive Power BI dashboard.
- Perform variance, trend, and bottleneck analysis.
- Identify likely drivers of delays and performance gaps.
- Recommend process improvements supported by metrics.
- Document requirements, assumptions, data definitions, risks, and limitations.
- Package the project as a reproducible GitHub portfolio project.

3. Scope

In scope

- Appointments and work orders
- Vehicle and customer information
- Service-center and technician performance
- Parts usage and costs
- Labor costs and service revenue
- Estimated versus actual completion time
- Cancellations and repeat visits
- Customer satisfaction
- Monthly and center-level reporting
- Process bottleneck analysis
- Power BI reporting and recommendations

Out of scope

- Real vehicle telemetry
- Real customer data
- Production deployment

- Payment gateway integration
- Mobile application development
- Machine-learning-based predictive maintenance

4. Technology Stack

Technology	Purpose	Priority
Python	Data generation, cleaning, validation, EDA	Required
Pandas / NumPy	Data manipulation and numerical analysis	Required
Matplotlib / Seaborn	Exploratory charts and analysis	Recommended
SQL	Joins, aggregation, business analysis, KPI queries	Required
PostgreSQL / MySQL	Relational data storage	Required
Power BI	Dashboard and management reporting	Required
DAX	Measures and KPI calculations	Required
Power Query	Data transformation	Recommended
Excel	Optional validation and management analysis	Optional
Git / GitHub	Version control and portfolio delivery	Recommended

5. Solution Architecture

Data Generation → Raw CSV → Python Cleaning/Validation → SQL Database → SQL Analysis → Power BI Model → DAX Measures → Dashboard → Findings → Process Recommendations

Keep the architecture deliberately simple. The goal is to demonstrate analytical, business, and process skills rather than cloud engineering.

6. Dataset Design

Generate realistic synthetic data. Never use real customer information. Start with a flat dataset if needed, then normalize it into relational tables.

Table	Key fields	Purpose
customers	customer_id, region, customer_type	Customer master
vehicles	vehicle_id, customer_id, model, year, mileage	Vehicle master
service_centers	center_id, city, region, capacity	Location/capacity
technicians	technician_id, center_id, skill_level	Technician master
appointments	appointment_id, vehicle_id, center_id, scheduled_date, status	Bookings
work_orders	work_order_id, appointment_id, technician_id, service_type, status, service_time	Actual service
parts	part_id, part_category, unit_cost	Parts master
part_usage	work_order_id, part_id, quantity	Parts consumed
financials	work_order_id, labor_cost, parts_cost, revenue	Financials
feedback	work_order_id, rating, feedback_category	Customer experience

7. Data Quality Requirements

- Check duplicate appointment and work-order IDs.
- Check missing foreign keys and required fields.
- Validate that completion time is not earlier than start time.
- Validate that costs and revenue are non-negative.
- Validate customer ratings are within 1–5.
- Validate date formats and date ranges.
- Check foreign-key relationships.
- Check for impossible mileage or model-year combinations.
- Document every cleaning rule.

Produce a data-quality report showing row counts, null counts, duplicates, invalid values, and final cleaned records.

8. SQL Requirements

SQL must demonstrate practical business analysis, not only basic SELECT statements.

- INNER JOIN and LEFT JOIN across service, vehicle, technician, parts, and financial tables.
- GROUP BY and aggregation for revenue, cost, service count, and turnaround time.
- CASE expressions for performance categories.
- CTEs for multi-step analysis.
- Window functions for technician and center rankings.
- Date functions for monthly and yearly trends.
- Views for reusable KPI datasets.

Required queries: monthly revenue/cost/profit; center turnaround time; on-time rate; technician workload; top service types; cancellation rate; customer rating by service type; highest-cost categories; estimated-versus-actual duration variance; repeat-visit rate.

9. Python Requirements

- Load raw data with Pandas.
- Inspect schema, nulls, duplicates, outliers, and distributions.
- Clean and standardize fields.
- Create turnaround_hours, cost_variance, profit, and on_time_flag.
- Perform descriptive statistics and trend analysis.
- Compare centers, technicians, and service types.
- Investigate relationships between workload, delays, costs, and satisfaction.
- Export cleaned datasets for SQL and Power BI.
- Document findings in a Jupyter notebook.

Do not claim causation from correlation alone. Treat observed relationships as hypotheses that require operational investigation.

10. KPI Framework

KPI	Calculation	Business question
Total Revenue	SUM(revenue)	How much revenue was generated?
Total Cost	labor + parts cost	What did servicing cost?

KPI	Calculation	Business question
Profit	revenue – total cost	Where is the operation financially efficient?
Average Service Cost	total cost / completed jobs	What does a typical service cost?
On-Time %	on-time jobs / completed jobs	How reliably are services completed?
Avg Turnaround	average actual duration	How long does service take?
Cost Variance	actual cost – estimated cost	Where are cost overruns?
Cancellation Rate	cancelled / appointments	Where is demand being lost?
Technician Utilization	service hours / available hours	How effectively is capacity used?
Customer Satisfaction	average rating	How do customers perceive service?
Repeat Visit Rate	returning customers / customers	Are customers coming back?

11. Power BI Dashboard

Page 1: Executive Overview

- Revenue, profit, cost, completed services, on-time %, turnaround time, customer rating.
- Monthly revenue and profit trend.
- Service-center performance ranking.
- Date, region, center, service-type, and model slicers.

Page 2: Operations

- Turnaround time by center.
- Technician workload/utilization.
- Estimated versus actual duration.
- Delay distribution.
- Service-type performance.

Page 3: Financial Analysis

- Revenue versus cost.
- Profit by center and service type.
- Cost variance.
- Parts versus labor cost.
- Monthly financial trend.

Page 4: Customer Experience

- Rating trend.
- Rating by service type and center.
- Cancellation rate.
- Repeat visits.
- Feedback categories.

12. Process Analysis

Map the end-to-end service workflow:

Customer Request → Appointment → Check-In → Inspection → Service Approval → Parts Allocation → Service → Quality Check → Payment → Handover → Feedback

For every stage, document the input, output, responsible role, expected duration, actual duration, failure points, and KPI.

Identify at least three bottlenecks from the data. Examples: excessive waiting time, parts-related delays, technician overload, rework, or appointment cancellations.

13. Root Cause Analysis

Use 5 Whys or a simple fishbone analysis. Example:

Problem: Center A has lower on-time completion. **Observation:** It also has high technician utilization and longer parts waiting times. **Investigation:** Compare workload, parts availability, service mix, and staffing with better-performing centers. **Recommendation:** Adjust technician allocation during peak periods and increase stock coverage for high-frequency parts.

Every recommendation must be traceable to an observed metric or pattern.

14. Requirements Engineering Deliverable

Create a Business Requirements Document containing:

- Business background and problem
- Project objectives
- Stakeholders
- Business requirements
- Functional requirements
- Non-functional requirements
- Data requirements
- KPI definitions
- Assumptions
- Constraints
- Risks
- Acceptance criteria

Example functional requirement: The dashboard shall allow service managers to filter KPIs by service center, date range, service type, and region.

Example acceptance criterion: Selecting a service center and date range must update all relevant KPIs and visualizations to show only the selected records.

15. Risk and Data Governance

- Use synthetic data only.
- Do not store personally identifiable customer information.
- Document KPI definitions and assumptions.
- Identify missing-data risks and their analytical impact.
- Use environment variables for database credentials.
- Never commit passwords or API keys.
- Document data lineage from source data to dashboard.

16. Recommended Repository Structure

```
vehicle-service-analytics/ ■■■ data/raw/ and data/cleaned/ ■■■ notebooks/01_data_generation.ipynb
■■■ notebooks/02_data_quality.ipynb ■■■ notebooks/03_exploratory_analysis.ipynb ■■■
sql/schema.sql ■■■ sql/data_quality.sql ■■■ sql/kpi_queries.sql ■■■ sql/views.sql ■■■
powerbi/vehicle_service_dashboard.pbix ■■■ docs/business_requirements.md ■■■ docs/process_map.md
■■■ docs/data_dictionary.md ■■■ docs/findings.md ■■■ docs/recommendations.md ■■■
src/data_cleaning.py ■■■ README.md ■■■ requirements.txt
```

17. Data Dictionary

Create a formal data dictionary for all important fields.

Field	Type	Example	Description
work_order_id	String	WO10234	Unique service work order
center_id	String	C012	Service-center identifier
vehicle_id	String	V08321	Unique vehicle
service_type	Category	Routine Service	Service performed
estimated_hours	Decimal	2.5	Expected duration
actual_hours	Decimal	3.2	Actual duration
labor_cost	Decimal	1800	Labor cost
parts_cost	Decimal	2450	Parts cost
revenue	Decimal	6200	Amount charged
customer_rating	Integer	4	Rating from 1 to 5
on_time_flag	Boolean	True	Whether target was met

18. Final Deliverables

Deliverable	Required content
Clean dataset	Validated data with documented cleaning rules
SQL database	Normalized tables, relationships, views, KPI queries
Python notebook	Cleaning, EDA, statistics, charts, findings
Power BI dashboard	Executive, operations, financial, customer pages
BRD	Requirements, stakeholders, assumptions, acceptance criteria
Process map	End-to-end workflow and bottlenecks
Data dictionary	Field definitions and business meaning
Findings report	Insights supported by metrics
Recommendations	Prioritized improvements with expected impact
README	Setup, architecture, screenshots, findings
GitHub repository	Clean and reproducible project

19. Minimum Acceptance Criteria

- At least 10,000 service records.
- Valid keys and documented relationships.
- Data-quality checks implemented.
- At least 10 meaningful SQL analytical queries.
- Python cleaning and exploratory analysis completed.
- At least 10 business KPIs.
- Power BI has at least 3 polished analytical pages.
- Dashboard filters work correctly.
- At least 3 operational bottlenecks identified.
- At least 3 data-supported recommendations.
- Requirements and process documentation included.

- No credentials or personal data in repository.
- Another person can reproduce the project from the README.

20. Optional Enhancements

- Demand forecasting.
- ABC analysis of parts consumption.
- Technician workload heatmap.
- What-if analysis for technician capacity.
- Power BI decomposition tree for profit drivers.
- Statistical testing between service centers.
- Automated Python data-quality report.

Do not add machine learning unless a clear business question justifies it. The core project is about BI, requirements, process analysis, and decision support.

21. Implementation Plan

Phase	Work	Output
1	Define problem and KPIs	Scope + KPI definitions
2	Design schema and generate data	Raw dataset + schema
3	Data-quality validation	Quality report
4	Python cleaning and EDA	Notebook + cleaned data
5	SQL database and analysis	Database + SQL scripts
6	Power BI model and DAX	Semantic model
7	Dashboard development	PBIX
8	Process and bottleneck analysis	Process map + findings
9	Requirements and recommendations	BRD + recommendations
10	Repository QA	Final GitHub project

22. Interview Demonstration

- Start with the business problem.
- Explain the dataset and relational model.
- Show one data-quality example.
- Demonstrate two or three SQL analyses.
- Show the Power BI executive dashboard.
- Drill into an underperforming service center.
- Explain the evidence behind one bottleneck.
- Present the recommended improvement.
- Explain how management would measure success.

Strong positioning: “I used operational data to identify a measurable business problem, investigated the drivers, and translated the findings into a process improvement recommendation.”

23. Definition of Done

The project is complete when a reviewer can understand the business problem, reproduce the data pipeline, inspect the SQL analysis, open the Power BI dashboard, trace KPIs back to documented data, and understand why the process

improvements were recommended.

The finished work should look like a small internal analytics engagement, not a generic college CRUD project.